Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains

A Review of Methods, Benchmarks, and Open Problems

Abstract

Large language models (LLMs) are increasingly consulted for scientific and biomedical information, yet their propensity to generate fluent but unsupported or incorrect statements — commonly termed hallucination — poses a substantial risk in high-stakes domains such as medicine, biology, and the physical sciences (Ji et al., 2023). This paper surveys the methods and benchmarks developed to measure the factual accuracy of LLMs across scientific domains, organizing the literature around three benchmark paradigms: multiple-choice knowledge testing (e.g., MMLU, GPQA), scientific claim verification against evidentiary corpora (e.g., SciFact, PubHealth, HealthVer, COVID-Fact, CLIMATE-FEVER), and open- or short-form factuality evaluation of generated text (e.g., TruthfulQA, SimpleQA, FActScore, FELM). We synthesize reported findings on model performance, describe current automated and human evaluation methodologies, and identify recurring limitations, including benchmark saturation, evidence contamination, domain narrowness, and the difficulty of evaluating long-form, compositional claims. We conclude by outlining research gaps — particularly the scarcity of benchmarks spanning the physical and life sciences beyond biomedicine, the lack of temporally updated scientific ground truth, and the need for calibrated uncertainty reporting — and propose directions for building more comprehensive, domain-representative, and dynamically maintained factuality benchmarks.

Keywords: large language models; factual accuracy; hallucination; benchmarking; scientific claim verification; biomedical natural language processing; evaluation methodology; retrieval-augmented generation

1. Introduction

Since the emergence of instruction-tuned large language models such as GPT-4, PaLM, and their successors, these systems have been rapidly adopted as general-purpose interfaces for scientific and technical information seeking, ranging from literature summarization to clinical decision support (Singhal et al., 2023). This adoption has occurred despite a well-documented tendency of LLMs to produce hallucinations — fluent, plausible-sounding text that is factually incorrect or unsupported by any verifiable source (Ji et al., 2023). In general-purpose conversational settings, a hallucinated fact may be an inconvenience; in scientific and biomedical contexts, however, factual errors can propagate into research summaries, clinical guidance, or public health communication, where the cost of misinformation is considerably higher.

Assessing the factual reliability of LLMs is therefore not merely an academic exercise but a prerequisite for responsible deployment. Over the past several years, the research community has produced a growing ecosystem of benchmarks intended to quantify LLM factuality, spanning general world knowledge (Lin, Hilton, & Evans, 2022; Wei et al., 2024), broad academic subject knowledge (Hendrycks et al., 2021), graduate-level scientific reasoning (Rein et al., 2023), and domain-specific scientific claim verification in biomedicine, public health, and climate science (Wadden et al., 2020; Kotonya & Toni, 2020; Diggelmann et al., 2020; Sarrouti et al., 2021; Saakyan, Chakrabarty, & Muresan, 2021).

This paper reviews the state of the art in benchmarking LLM factual accuracy specifically as it pertains to scientific domains. We synthesize evidence from claim-verification datasets, knowledge-probing multiple-choice benchmarks, and long-form factuality evaluation frameworks, and we examine both the automated and human evaluation methodologies used to score model outputs. We further discuss the structural challenges that make scientific factuality especially difficult to benchmark — including the compositional and evidence-dependent nature of scientific claims, the rapid obsolescence of scientific consensus, and the scarcity of benchmarks outside biomedicine — before outlining directions for future benchmark design.

2. Background and Related Work

The study of factuality in natural language generation predates the current generation of LLMs, with early work on hallucination concentrating on abstractive summarization and data-to-text generation, where models were shown to introduce content unsupported by the source (Ji et al., 2023). Ji et al. (2023) provide a comprehensive taxonomy distinguishing intrinsic hallucination, in which generated content directly contradicts the source material, from extrinsic hallucination, in which content is unverifiable against the source and may originate from parametric knowledge learned during pretraining. This distinction remains central to scientific-domain benchmarking, since a model may misstate a study's own findings (intrinsic error) or assert an unrelated but false scientific claim without any grounding text at all (extrinsic error).

A second strand of related work concerns general-purpose factuality and knowledge benchmarks. TruthfulQA (Lin, Hilton, & Evans, 2022) introduced 817 adversarially constructed questions spanning 38 categories designed to elicit imitative falsehoods — answers that mimic common human misconceptions. The authors reported that their best-performing model at the time was truthful on only 58% of questions compared to 94% for humans, and, notably, that larger models were not more truthful under the default prompting condition, suggesting that scale alone does not resolve factuality. Building on this line of work, OpenAI's SimpleQA (Wei et al., 2024) offered a more contemporary, adversarially collected set of short, fact-seeking questions with single indisputable answers, explicitly designed to remain discriminating for frontier models that had begun to saturate earlier benchmarks such as TriviaQA and Natural Questions.

A third strand, more directly relevant to this review, concerns scientific and biomedical knowledge and claim verification. The Massive Multitask Language Understanding benchmark (MMLU; Hendrycks et al., 2021) assesses multiple-choice knowledge across 57 subjects, many of which are scientific (e.g., anatomy, college biology, college chemistry, college physics, clinical knowledge), and has become a standard reference point for reporting LLM capability, though it has since become saturated by frontier models. GPQA (Rein et al., 2023) addresses this saturation by presenting 448 graduate-level, expert-written multiple-choice questions in biology, physics, and chemistry that are deliberately "Google-proof": domain PhD holders reached 65% accuracy, non-expert validators with unrestricted web access reached only 34%, and a GPT-4-based baseline achieved 39% accuracy, illustrating a substantial gap between frontier-model performance and expert human performance on genuinely difficult scientific reasoning.

In parallel, a body of claim-verification datasets has emerged specifically to test whether models — or retrieval-augmented pipelines built around them — can correctly judge whether a scientific claim is supported, refuted, or unverifiable given a corpus of evidence. SciFact (Wadden et al., 2020) is foundational to this literature, comprising 1,409 expert-written biomedical claims paired with a corpus of 5,183 paper abstracts, each annotated with SUPPORTS, REFUTES, or NOT ENOUGH INFO (NEI) labels and supporting rationale sentences. SciFact was later extended by SciFact-Open (Wadden et al., 2022), which scales the evidence corpus to roughly 500,000 abstracts to create a more realistic open-domain retrieval setting. Related biomedical and public-health claim datasets include PubHealth (Kotonya & Toni, 2020), which draws claims from journalists and fact-checkers; HealthVer (Sarrouti et al., 2021), which sources claims from real-world COVID-19-related search queries verified against PubMed literature; and COVID-Fact (Saakyan, Chakrabarty, & Muresan, 2021), which extracts claims and counterclaims from social media discussion of the pandemic. Outside biomedicine, CLIMATE-FEVER (Diggelmann et al., 2020) applies the same claim-verification paradigm to real-world claims about climate science drawn from news media.

3. Dimensions of Factual Accuracy Benchmarking Across Scientific Domains

3.1 Defining Factuality and Hallucination in Scientific Contexts

Factuality in the context of LLMs is generally operationalized as the degree to which generated content is consistent with an external, verifiable ground truth, whether that ground truth is world knowledge, a cited source document, or established scientific consensus (Ji et al., 2023). In scientific domains this operationalization is complicated by the compositional and probabilistic nature of scientific claims: a statement about drug efficacy or a physical constant may be conditionally true (dependent on population, methodology, or measurement context), may be actively contested in the literature, or may become outdated as new evidence accumulates. Benchmarks such as SciFact address this by defining a claim as verifiable only relative to a specific evidentiary abstract, explicitly separating the question "is this claim true" from "is this claim supported by this document," the latter being a more tractable and reproducible annotation target (Wadden et al., 2020).

3.2 A Domain Taxonomy of Scientific Factuality Benchmarks

Scientific-domain benchmarks can be organized along two axes: the scientific subdomain covered and the task format used to elicit and score factual behavior. On the subdomain axis, biomedicine and clinical medicine are by far the most extensively benchmarked areas, reflecting both the high stakes of medical misinformation and the relative availability of curated biomedical corpora such as PubMed abstracts. PubMedQA (Jin et al., 2019) exemplifies this concentration: it comprises 1,000 expert-labeled instances requiring yes/no/maybe answers to research questions grounded in PubMed abstracts, plus substantially larger machine-labeled and unlabeled subsets, and the original paper reported that even a fine-tuned BioBERT model reached only 68.1% accuracy against a single-human performance ceiling of 78.0%, underscoring that reasoning over quantitative biomedical findings remains difficult even for specialized systems. BioASQ (Krithara, Nentidis, Bougiatiotis, & Paliouras, 2023) similarly provides a manually curated, expert-annotated corpus for biomedical question answering that has been used for over a decade as a shared-task benchmark. By contrast, the physical and mathematical sciences are comparatively underrepresented in dedicated factuality benchmarks; GPQA (Rein et al., 2023) is one of the few resources offering graduate-level, expert-authored factual questions in biology, physics, and chemistry jointly, while MMLU (Hendrycks et al., 2021) offers broader but shallower multiple-choice coverage across many scientific subjects.

On the task-format axis, three broad paradigms recur across the literature. The first is multiple-choice knowledge probing, exemplified by MMLU and GPQA, in which a model selects among a small set of answer options; this format is easy to score automatically but conflates factual recall with test-taking heuristics and is vulnerable to memorization of benchmark items during pretraining. The second is evidence-grounded claim verification, exemplified by SciFact, PubHealth, HealthVer, COVID-Fact, and CLIMATE-FEVER, in which a system must classify a claim as supported, refuted, or unverifiable relative to a retrieved or provided evidence set, typically also identifying supporting rationale sentences (Wadden et al., 2020; Kotonya & Toni, 2020). This format more directly tests the kind of evidence-based reasoning required of a system used for literature review or fact-checking, but requires substantial annotation effort and is generally restricted to abstract-level rather than full-text evidence. The third paradigm is open- or long-form factuality evaluation, in which a model is asked to generate free text (for example, a biography, an explanation, or an answer to an open-ended question) and the output is decomposed into atomic factual claims that are individually verified. FActScore (Min et al., 2023) formalizes this approach for biographical generation, reporting that ChatGPT-generated biographies achieved only 58% atomic factual precision against a reliable knowledge source, despite being judged more factual than open, non-commercial models of the time such as Alpaca and Vicuna. FELM (Chen et al., 2023) extends fine-grained factuality annotation beyond world knowledge to additional domains including mathematics and reasoning, explicitly motivated by the observation that prior factuality evaluators concentrated too narrowly on Wikipedia-style world knowledge.

3.3 Short-Form Parametric Factuality

A complementary line of work targets short-form, single-answer factuality independent of retrieval or provided evidence, probing what a model "knows" from its parametric memory alone. TruthfulQA (Lin, Hilton, & Evans, 2022) was among the first such benchmarks, focused on questions where a plausible but false answer mirrors a common human misconception; several of its 38 categories touch directly on science, health, and misconceptions about biology and physics. SimpleQA (Wei et al., 2024) subsequently offered a broader, adversarially collected set of over four thousand short factual questions spanning science, history, geography, and popular culture, explicitly designed to remain difficult for frontier models and to reward calibrated abstention — that is, a well-behaved model should decline to answer questions it is not confident about rather than confabulate a plausible-sounding but incorrect response. This attempt-versus-abstain framing is particularly relevant to scientific use cases, where an overconfident incorrect answer is generally more harmful than an explicit statement of uncertainty.

4. Current Approaches and Developments

Beyond the underlying benchmark datasets, the research community has converged on several methodological approaches for scoring LLM factuality, each with distinct trade-offs between fidelity, cost, and scalability.

4.1 Human Evaluation Frameworks

Human annotation remains the gold standard for scientific factuality assessment, particularly for long-form or clinically consequential text. Singhal et al. (2023) introduced MultiMedQA, a benchmark combining six existing medical question-answering datasets — including MedQA, MedMCQA, and PubMedQA — with a newly collected consumer health question set, HealthSearchQA, and proposed a multi-axis human evaluation rubric assessing factuality, comprehension, reasoning, potential harm, and bias in model-generated answers. This multi-axis approach reflects a broader recognition that a single factuality score is insufficient in clinical contexts, where an answer can be simultaneously factually accurate yet potentially harmful if incomplete or poorly calibrated to a lay audience.

4.2 Automated and LLM-as-Judge Evaluation

Because human evaluation is costly and slow, automated factuality scoring has become increasingly central to benchmark development. FActScore (Min et al., 2023) demonstrated that an automated estimator combining retrieval with a strong language model could approximate human-judged factual precision with under 2% error, enabling the evaluation of 6,500 generations from thirteen language models at a fraction of the cost of full human annotation.

FELM (Chen et al., 2023) similarly investigates the reliability of LLM-based factuality evaluators themselves, finding that retrieval augmentation improves evaluator performance but that current LLM evaluators still fall short of reliably verifying fine-grained factual claims across diverse domains, an important caution against treating LLM-as-judge pipelines as a fully trustworthy substitute for human review.

4.3 Retrieval-Augmented and Evidence-Grounded Approaches

Retrieval-augmented generation (RAG) is widely used both as a mitigation strategy for hallucination and as a component of the evaluation pipelines themselves. In the SciFact task formulation, systems are explicitly evaluated on their ability to retrieve relevant abstracts and correctly classify claims against them (Wadden et al., 2020), and the later SciFact-Open extension (Wadden et al., 2022) increases the evidence corpus from roughly five thousand to five hundred thousand abstracts specifically to test whether retrieval-based pipelines degrade under more realistic open-domain conditions. Domain-specific biomedical RAG frameworks similarly draw on PubMedQA, MedMCQA, and BioASQ as retrieval and evaluation corpora, allowing model outputs to be cross-validated against primary medical literature rather than relying solely on parametric knowledge.

4.4 Observed Performance Patterns

Across these benchmarks, several consistent patterns emerge from the reported literature. First, model performance on scientific factuality tasks has generally improved with newer model generations, but a persistent gap relative to domain-expert human performance remains on the most difficult items — for example, GPT-4-based systems reached only 39% accuracy on GPQA against a 65% expert human baseline (Rein et al., 2023). Second, scaling model size alone does not reliably improve truthfulness on adversarially constructed questions; Lin, Hilton, and Evans (2022) found that larger models were, if anything, less truthful under naive prompting on TruthfulQA, since larger models are better at reproducing common human misconceptions learned from training text. Third, atomic, fine-grained scoring methods such as FActScore reveal substantially lower factual precision in long-form generation than aggregate or holistic human judgments might suggest, with even strong commercial models such as ChatGPT scoring around 58% on biographical generation (Min et al., 2023). Together these findings suggest that headline capability improvements on aggregate benchmarks do not necessarily translate into proportionate improvements in fine-grained scientific factuality.

5. Research Challenges and Limitations

Despite substantial progress, several structural challenges limit the reliability and generalizability of current factuality benchmarks in scientific domains.

Benchmark saturation and contamination. Widely used benchmarks such as MMLU and early versions of TriviaQA and Natural Questions have become saturated by frontier models, in part because benchmark items or closely related content may appear in pretraining corpora, and in part because models are increasingly optimized against these exact evaluation sets (Wei et al., 2024). This motivates continual benchmark refreshes, but also means that longitudinal comparisons across model generations are of uncertain validity once contamination is suspected.

Domain imbalance. The literature on scientific claim verification and factuality benchmarking is heavily concentrated in biomedicine and, to a lesser extent, public health and climate science (Wadden et al., 2020; Kotonya & Toni, 2020; Sarrouti et al., 2021; Diggelmann et al., 2020). Benchmarks jointly covering the physical sciences, chemistry, and mathematics at a comparable scale and evidentiary rigor remain comparatively scarce; GPQA (Rein et al., 2023) is a notable but small-scale exception, containing only 448 items across three disciplines. This imbalance limits the field's ability to characterize LLM factuality across the full breadth of scientific inquiry.

Evidence granularity and full-text access. Most claim-verification datasets, including SciFact and its extensions, operate at the level of paper abstracts rather than full text, since abstracts are more readily available and easier to annotate exhaustively (Wadden et al., 2022). This restricts the evaluation of claims whose supporting or contradicting evidence lies in a paper's methods, results, or discussion sections, and may understate the retrieval difficulty faced by systems used against real scientific literature.

The changing nature of scientific ground truth. Unlike encyclopedic world knowledge, scientific consensus evolves as new evidence accumulates, and a claim considered well-supported at benchmark construction time may later be revised or overturned. Static benchmarks such as SciFact and PubHealth necessarily reflect a snapshot of the literature at construction time (Wadden et al., 2020; Kotonya & Toni, 2020), and none of the reviewed benchmarks incorporate a mechanism for systematically updating ground-truth labels as scientific understanding changes.

Automated evaluator reliability. Although automated and LLM-as-judge scoring methods substantially reduce the cost of factuality evaluation, their own reliability is imperfect and domain-dependent. Chen et al. (2023) found that current LLM-based factuality evaluators, even when augmented with retrieval, do not yet reliably assess factuality across the diverse domains spanning world knowledge, mathematics, and reasoning that their FELM benchmark covers, indicating that the evaluators themselves require independent validation rather than being treated as ground truth.

Compositional and quantitative reasoning. Several biomedical benchmarks explicitly note that correctly answering their questions requires reasoning over quantitative content in source abstracts rather than simple lexical matching (Jin et al., 2019). Fine-grained factuality errors involving numerical values, statistical significance, or dosage information are more consequential in scientific and clinical settings than in general-domain factuality tasks, yet most existing atomic-fact-decomposition methods, including FActScore, were developed and validated primarily on biographical or encyclopedic text rather than quantitative scientific claims (Min et al., 2023).

6. Research Gaps and Future Directions

Building on the challenges identified above, we highlight several concrete directions for future benchmark development and evaluation methodology.

Broader and more balanced domain coverage. Future benchmarking efforts should extend the claim-verification paradigm established by SciFact and its biomedical counterparts to a wider range of scientific subdomains, including chemistry, physics, earth science, and engineering, ideally with comparable scale, expert annotation quality, and evidence granularity. GPQA (Rein et al., 2023) demonstrates that expert-authored, difficulty-calibrated items are feasible to construct even for a small benchmark; scaling this approach while retaining evidentiary grounding, in the manner of SciFact, is a promising but resource-intensive direction.

Full-text and multi-document evidence. Extending evidence corpora beyond abstracts to full-text scientific articles, as partially explored by the open-domain retrieval setting in SciFact-Open (Wadden et al., 2022), would better reflect the retrieval and synthesis demands placed on LLM-based research assistants, which increasingly need to reconcile evidence across multiple, sometimes conflicting, full-length publications.

Dynamic and continually updated benchmarks. Because scientific consensus evolves, static benchmarks risk becoming stale or, worse, penalizing models for reflecting more current evidence than the benchmark's original annotation. Benchmark maintenance protocols that periodically re-verify claim labels against the current literature, analogous in spirit to continually refreshed general-knowledge benchmarks, represent an underexplored but important direction for scientific factuality evaluation.

Calibration and abstention as first-class metrics. Following the attempt-versus-abstain framing introduced by SimpleQA (Wei et al., 2024), future scientific factuality benchmarks should more systematically reward calibrated uncertainty, since an LLM that appropriately declines to answer a scientific question outside its reliable knowledge is preferable, in high-stakes domains, to one that confabulates a confident but incorrect response.

Validated automated evaluators for scientific text. Given the imperfect reliability of current LLM-as-judge factuality evaluators (Chen et al., 2023), further work is needed to validate automated scoring methods specifically against expert-annotated scientific and quantitative claims, rather than relying on evaluator performance profiles established primarily on general-domain or biographical text (Min et al., 2023).

Multimodal and structured scientific content. Much scientific evidence is conveyed through figures, tables, and structured data rather than prose alone. Extending factuality benchmarking to require verification against tables and figures, building on early work in this direction, would better reflect the full evidentiary basis of scientific claims and address a currently underserved evaluation gap.

7. Conclusion

The factual reliability of large language models in scientific domains is now supported by a substantial and rapidly growing body of benchmarks, spanning multiple-choice knowledge probing, evidence-grounded claim verification, and fine-grained long-form factuality evaluation. Collectively, this literature demonstrates measurable progress in model capability alongside persistent and well-documented failure modes: performance gaps relative to domain experts on the hardest scientific reasoning tasks (Rein et al., 2023), unreliable truthfulness gains from scale alone (Lin, Hilton, & Evans, 2022), and substantially lower fine-grained factual precision than aggregate benchmarks might suggest (Min et al., 2023). The benchmarking ecosystem remains heavily weighted toward biomedicine, relies predominantly on abstract-level rather than full-text evidence, and lacks mechanisms for keeping pace with evolving scientific consensus. Addressing these gaps — through broader domain coverage, richer evidence grounding, dynamic benchmark maintenance, and better-validated automated evaluators — represents a necessary next step toward LLMs that can be responsibly relied upon for scientific and biomedical information tasks.

References

Chen, S., Zhao, Y., Zhang, J., Chern, I.-C., Gao, S., Liu, P., & He, J. (2023). FELM: Benchmarking factuality evaluation of large language models. arXiv preprint. arXiv:2310.00741. https://arxiv.org/abs/2310.00741

Diggelmann, T., Boyd-Graber, J., Bulian, J., Ciaramita, M., & Leippold, M. (2020). CLIMATE-FEVER: A dataset for verification of real-world climate claims. arXiv preprint. arXiv:2012.00614. https://arxiv.org/abs/2012.00614

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., & Steinhardt, J. (2021). Measuring massive multitask language understanding. In Proceedings of the 9th International Conference on Learning Representations (ICLR 2021). arXiv:2009.03300. https://arxiv.org/abs/2009.03300

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. ACM Computing Surveys, 55(12), Article 248, 1–38. https://doi.org/10.1145/3571730

Jin, Q., Dhingra, B., Liu, Z., Cohen, W., & Lu, X. (2019). PubMedQA: A dataset for biomedical research question answering. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), 2567–2577. arXiv:1909.06146. https://arxiv.org/abs/1909.06146

Kotonya, N., & Toni, F. (2020). Explainable automated fact-checking for public health claims. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 7740–7754. Association for Computational Linguistics.

Krithara, A., Nentidis, A., Bougiatiotis, K., & Paliouras, G. (2023). BioASQ-QA: A manually curated corpus for biomedical question answering. Scientific Data, 10(1), 170. https://doi.org/10.1038/s41597-023-02068-4

Lin, S., Hilton, J., & Evans, O. (2022). TruthfulQA: Measuring how models mimic human falsehoods. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 3214–3252. https://doi.org/10.18653/v1/2022.acl-long.229 (arXiv:2109.07958)

Min, S., Krishna, K., Lyu, X., Lewis, M., Yih, W., Koh, P. W., Iyyer, M., Zettlemoyer, L., & Hajishirzi, H. (2023). FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP), 12076–12100. https://doi.org/10.18653/v1/2023.emnlp-main.741 (arXiv:2305.14251)

Rein, D., Hou, B. L., Stickland, A. C., Petty, J., Pang, R. Y., Dirani, J., Michael, J., & Bowman, S. R. (2023). GPQA: A graduate-level Google-proof Q&A benchmark. arXiv preprint. arXiv:2311.12022. https://doi.org/10.48550/arXiv.2311.12022

Saakyan, A., Chakrabarty, T., & Muresan, S. (2021). COVID-Fact: Fact extraction and verification of real-world claims on COVID-19 pandemic. In Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP), 2116–2129.

Sarrouti, M., Abacha, A. B., M'rabet, Y., & Demner-Fushman, D. (2021). Evidence-based fact-checking of health-related claims. In Findings of the Association for Computational Linguistics: EMNLP 2021, 3499–3512.

Singhal, K., Azizi, S., Tu, T., Mahdavi, S. S., Wei, J., Chung, H. W., Scales, N., Tanwani, A., Cole-Lewis, H., Pfohl, S., Payne, P., Seneviratne, M., Gamble, P., Kelly, C., Babiker, A., Schärli, N., Chowdhery, A., Mansfield, P., Demner-Fushman, D., ... Natarajan, V. (2023). Large language models encode clinical knowledge. Nature, 620(7972), 172–180. https://doi.org/10.1038/s41586-023-06291-2

Wadden, D., Lin, S., Lo, K., Wang, L. L., van Zuylen, M., Cohan, A., & Hajishirzi, H. (2020). Fact or fiction: Verifying scientific claims. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 7534–7550. Association for Computational Linguistics.

Wadden, D., Lo, K., Kuehl, B., Cohan, A., Beltagy, I., Wang, L. L., & Hajishirzi, H. (2022). SciFact-open: Towards open-domain scientific claim verification. arXiv preprint. arXiv:2210.13777. https://arxiv.org/abs/2210.13777

Wei, J., Karina, N., Chung, H. W., Jiao, Y. J., Papay, S., Glaese, A., Schulman, J., & Fedus, W. (2024). Measuring short-form factuality in large language models. arXiv preprint. arXiv:2411.04368. https://arxiv.org/abs/2411.04368